Paleomagnetism of the Nipissing diabase, Cobalt area, Ontario

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At Cobalt the Nipissing diabase intrusive occurs as a 1000 ft (~305 m) thick undulating sheet exposed in a series of basin and arch structures from which 112 specimens were collected at 14 sites. Paleomagnetic analysis of the cleaned natural remanence directions from the 11 acceptable sites suggests that these structures are dominantly primary. Minor post-solidification compression associated with the NW–SE trending faults may have amplified the primary NE–SW axial trends of the structures. Significant folding adjacent to the faults does not appear probable. The origin of the narrow argentiferous veins is probably related to minor local differential movements of large blocks of diabase or to cooling-contraction of the sheet. A revised pole position of 91.9 °E, 19.4 °N (dp = 3.9°; dm = 6.5°) is given for the 2140 m.y. old Nipissing diabase.

Introduction

In the Cobalt area of Ontario, the Nipissing diabase intrudes steeply-dipping Archean Keewatin volcanics and gently-dipping Cobalt Group sediments which overlie the volcanics with angular unconformity. The diabase is an undulating sill-like sheet up to 1000 ft (~305 m) thick with a basin and arch structure (Thompson 1961). It has been dated at 2180 ± 40 m.y. by the Rb–Sr whole rock method (Van Schmus 1965) and at 2095 m.y. by the K–A whole rock method (Lowdon *et al.* 1963). Miller (1910) considered the diabase structure to be primary (i.e. to be formed prior to solidification of the diabase) except for nearly vertical fault offsets. Knight (1922) pointed to the gentle dips in the Cobalt Group and concluded that post-solidification folding of the diabase was at most very minor except adjacent to the major NE and NW trending faults. Thomson (1957) considered the basin and arch structure to be dominantly primary with some secondary or post-solidification deformation being important in the genesis of the associated silver–cobalt vein deposits. Noting spatial and mineralogic relationships, Moore (1934) concluded that the silver–cobalt mineralization emanated from the diabase magma. Thomson (1957) observed that all of the important silver veins occurred in nearly vertical veins which bottom in sedimentary Keewatin interflow bands containing sulfide mineralization, developed munificently in the overlying Cobalt Group sediments, and pinched out in the uppermost Nipissing diabase. After mineralogic studies, Petruk (1968) suggested that the silver–cobalt minerals were remobilized by heat from the diabase intrusion, and migrated out of the sulfide zone up preexisting fractures which followed the structurally weaker interflow bands.

From a previous paleomagnetic study (Symons 1967), the Nipissing diabase was known to carry a stable primary natural remanent magnetization (n.r.m.) of probable thermoremanent magnetization (t.r.m.) origin residing in titanomagnetite. Subsequent petrologic studies by Christoffersen (1968) confirmed that titanomagnetite is the major magnetic mineral in the diabase. This study is concerned with analyzing remanence directions from the Nipissing diabase to examine the origin of its structure. For example, if the basin and arch structure is primary (secondary), then the remanence directions of variously dipping limbs should disperse (coalesce) when the limbs are unfolded to a horizontal plane (Graham 1949).

Procedure

Four drill cores, oriented *in situ* by sun compass, were collected at each of 14 sites in the Nipissing diabase (Fig. 1). Two specimens, each 1¼ in. (~3.2 cm) in diameter and 1⅛ in. (~2.9 cm) in height, were cut from each core and their n.r.m. measured using an automated biastatic magnetometer (Larochelle and Christie 1967). One pilot specimen per site with an "average" n.r.m. direction and intensity was stepwise demagnetized in alternating fields (a.f.) of 50, 100, 200, 300, 400, and 800 oe (peak) using the apparatus described by Larochelle

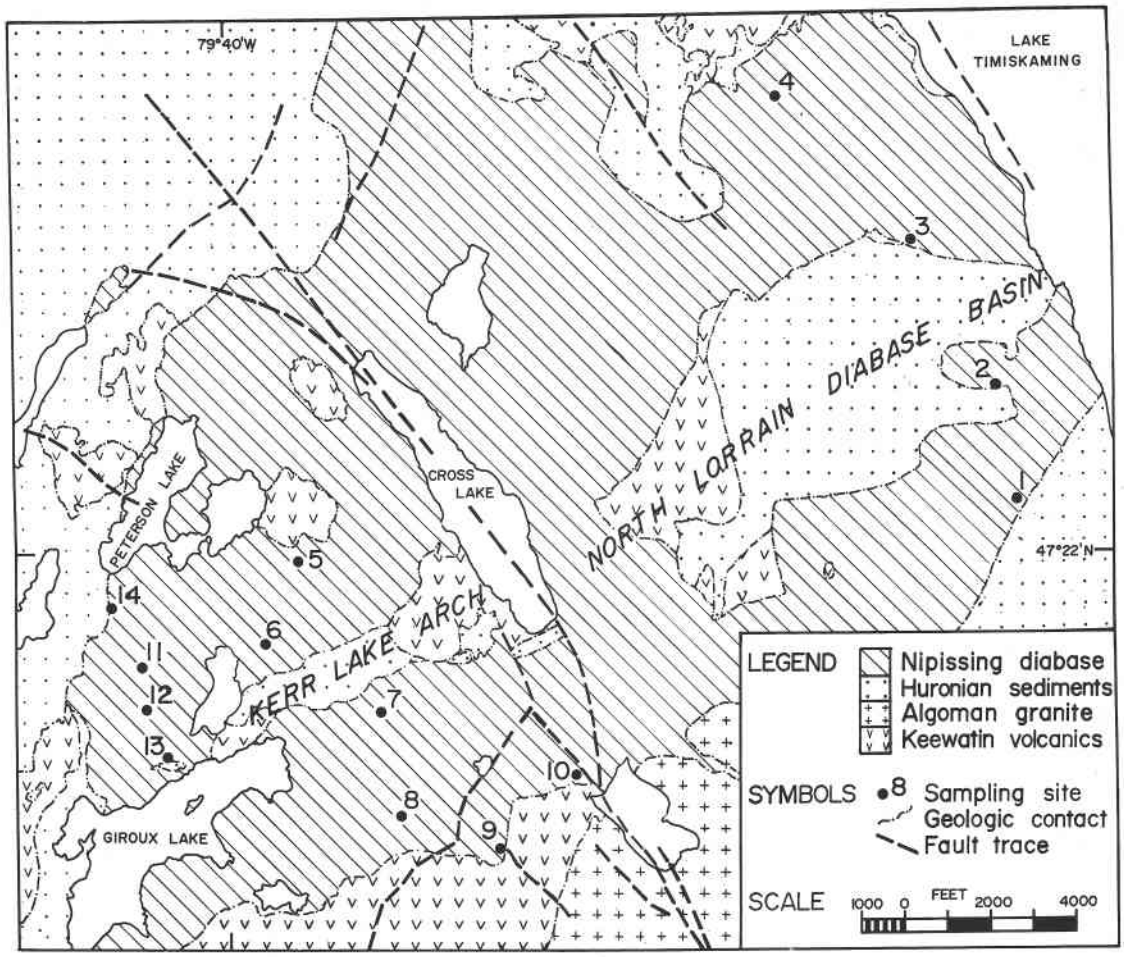


FIG. 1. Location of paleomagnetic sampling sites. Geology from Ontario Dept. Mines Maps 2050 and 2052, Cobalt Silver Area, Northern and Southeastern Sheets by R. Thomson, 1964.

and Black (1965). After examining the behavior of the remanence of the pilot specimens during stepwise demagnetization and after calculating their stability index (Tarling and Symons 1967), an optimum demagnetizing field was selected for each site as noted in Table I in which its remaining specimens were demagnetized.

The reliability of the remanence data was examined on the following basis:

(*1*) If the angle between the two specimen remanence vectors from one core exceeded 23°, which corresponds to a maximum core mean standard error of 11.5°, then the core magnetization was not considered sufficiently homogeneous to be reliable (Larochelle 1968). Nine cores were rejected on this basis with 2, 2, 3, 1, and 1 cores from sites 5, 6, 8, 11, and 13 respectively.

(*2*) At site 8, only one reliable core remained. It was rejected because the site mean standard error could not be calculated.

The mean remanence vector was calculated for each homogeneously magnetized core and used to calculate the mean site remanence direction and spherical statistics (Larochelle 1968).

(*3*) Sites 9 and 10 each had one core with an anomalous remanence direction which deviated by more than 70° from the mean direction or by more than three times the within-site standard deviation of the other 3 cores. The anomalous cores were rejected and the site statistics recalculated.

(*4*) Sites 5 and 6 each had 2 reliable cores;

TABLE I

Summary of site remanence measurements

SITE*	STR†	DIP†	N‡	H_{ac}§	Demagnetized remanence DEC‖	INC‖	R¶	SE**	α_{95}
1	215	12W	4	300	9.1	−41.0	3.967	4.3	7.4
2	215	12W	4	100	354.6	−40.0	3.987	2.7	4.6
3	70	12S	4	200	2.4	−42.5	3.872	8.3	14.5
4	70	12S	4	300	21.2	−50.6	3.700	12.8	22.2
5	240	10N		300					
6	240	30N		300					
7	60	15S	4	300	16.4	−51.0	3.981	3.3	5.6
8	60	15S		300					
9	10	60E	3	400	6.5	−34.2	2.994	6.8	13.5
10	80	15S	3	200	358.8	−21.3	2.930	7.5	15.1
11	0	0	3	200	8.6	−41.6	2.990	1.1	1.9
12	0	0	4	200	12.8	−42.3	3.997	1.3	2.2
13	0	0	3	200	11.7	−31.1	2.987	3.1	6.3
14	0	0	4	300	18.0	−40.5	3.996	1.5	2.6
Mean			11††		8.6	−39.9	10.834	3.1	5.4

*SITE = Sampling site number.
†STR, DIP = Strike and dip of diabase sheet in degrees.
‡N = Number of cores included in statistics of the 4 collected.
§H_{ac} = Peak intensity of alternating field demagnetization in oersteds.
‖DEC, INC = Declination and inclination of mean remanence direction in degrees.
¶R = Length of resultant of mean vector giving each core unit weight.
**SE, α_{95} = Radius of standard error and cone of 95% confidence of mean remanence direction in degrees.
†† = Number of sites.

however, they yielded α_{95} values of 36° and 62° respectively and mean site standard errors in excess of 11.5°. Both sites were considered unreliable for this study and rejected from further consideration.

Discussion

Eleven of the 14 sites sampled were considered to have reliable mean-site remanence directions. The strike and dip of the diabase sheet at the 11 sites (Table I) were obtained from information compiled by Thomson (1960, pp. 18–20; 1961*a*, pp. 46–47; 1961*b*, pp. 17, 19, 22, 24, 52; 1961*c*, p. 43). All tectonic corrections were made by simple rotation of the dip plane to the horizontal about the strike. Applying the fold test (Graham 1949) to all 11 mean site remanence directions, the values of α_{95} before and after tectonic correction were 5.4° and 8.6° respectively. Snedecor's F-ratio test for the null hypothesis (Larochelle 1968) indicates that there is a probability of 0.97 that the scatter in the remanence directions caused by applying tectonic corrections is significant. These tests show that the diabase acquired its remanence after solidification of the primary basin and arch structure.

There is some evidence that Thomson's (1957) suggestion that post-solidification tectonics accentuated the basin and arch structure is correct. The 8 cores from sites 1 and 2 from the south limb of the Lorrain Diabase Basin yield a mean remanence direction of 1.8°, −40.7° (α_{95} = 5.4°) whereas the 8 cores from sites 3 and 4 on the north limb yield 10.9°, −46.8° (α_{95} = 13.2°). Despite the fact that the F-ratio test is inconclusive, it is noteworthy that the remanence directions are brought into best agreement if tectonic corrections of 6° instead of the actual 12° are applied to each limb using a strike of N 40°E. Such a correction is consistent with some post-solidification NW–SE contraction approximately perpendicular to the basin axis (N 50°E) and parallel to the major NW–SE fault set. Unfortunately 3 of the 4 sites (5, 6, 7, and 8) collected to define the limbs of the Kerr Lake Arch which might have provided useful additional data, are not reliable. Thus the data favor a minor amount of secondary folding but not on a statistical significance basis.

Sites 9 and 10 are adjacent to the Beaver and Kerr Lake faults respectively where the diabase sheet dips at 60° and 15° respectively. Application of tectonic corrections causes the remanence directions to scatter from consistent primary directions to completely inconsistent directions. This result suggests that the diabase

was not folded by more than a very few degrees adjacent to the major faults contrary to Knight's (1922) suggestion.

Sites 11 to 14 are collected on a N–S traverse perpendicular to the E–W ore vein structures of the Silverfields mine where the diabase sheet is flat-lying (Moore 1967; Petruk 1968). If post-solidification tectonics caused the sheet to fracture into blocks defined by the veins, then the remanence directions might reflect some consistent deformation pattern. The circles of 95% confidence of adjacent sites do not intersect so that the sites do have statistically significantly different remanence directions with intersite differences of 10°, 4°, and 7°. However, there is no apparent consistent pattern to the movement between sites suggesting that the variations reflect either a few degrees of random tilting or rotation of the blocks, or a few degrees of secular variation of the ambient magnetic field during cooling of the sill.

The dipole formula is used to calculate a pole position of 91.9 °E, 19.4 °N (dp = 3.9°, dm = 6.5°)[1] for the Nipissing diabase from the 11 mean site remanence directions. This position is considered more reliable than the previously reported position (74.4 °E, 11.9 °N, dp = 5.7°, dm = 9.0°; Symons 1967) because of its broader sampling base, improved technology and smaller oval of 95% confidence. More specifically, site 12 is over the Silverfields Mine site of the previous sampling and their remanence directions differ by only 12°. The 18° shift in the Nipissing diabase pole position does not alter the paleomagnetic interpretation given in the previous paper of the geological sequence of events in the Cobalt area, of the behavior of the earth's magnetic field during the Aphebian, or of the magnetic properties of the rocks.

Conclusions

From this paleomagnetic study of the Nipissing diabase it is concluded that: (*1*) the basin and arch structure of the diabase sheet is undoubtedly dominantly primary; (*2*) minor post-solidification NW–SE contraction possibly occurred parallel to the NW–SE fault set and perpendicular to the NE–SW axes of the basins and arches; (*3*) folding of the diabase adjacent to the faults was apparently slight if present at all; (*4*) formation of the silver–cobalt vein structures in the diabase can be explained by either cooling contraction of the diabase sheet, or by minor post-solidification tilting or rotation (block faulting) of sections of the sheet; and (*5*) the updated location of the pole position for the Nipissing diabase (age about 2140 ± 60 m.y.) is 91.9 °E, 19.4 °N (dp = 3.9°, dm = 6.5°).

[1]Fisher's (1953) statistics give (dp = 4.0°, dm = 6.6°).

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